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Reverse design of molecule-process-process networks: A case study from HEN-ORC system

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Abstract

The integrated design of the heat exchanger network (HEN) and organic Rankine cycle (ORC) system with new working fluids is a complex optimization problem. It involves navigating a vast design space across working fluid molecules, ORC processes, and networks. In this article, a new two-stage reverse strategy is developed. The optimal HEN-ORC configurations and operating conditions, and the thermodynamic properties of the hypothetical working fluid are identified by an equation of state (EOS) free HEN-ORC model in the first stage. With two developed group contribution-artificial neural network thermodynamic property prediction models, working fluid molecules are screened out in the second stage from a database containing more than 430,000 hydrofluoroolefins (HFOs). The presented method is employed in two cases, where new working fluids are found. The total annual cost of Case 1 is 12%–22% lower than the literature, and the power output of Case 2 is 5%–8% higher than the literature.

KEYWORDS
ANN, HEN-ORC, hydrofluoroolefin, reverse design

1 | INTRODUCTION

Non-renewable energy (e.g., oil, natural gas, and coal) accounts for more than 80% of global energy consumption, leading to a large number of carbon emissions. About 40%–60% of the energy in the industry becomes waste heat and is directly discharged into the environment. Since more than 50% of industrial waste heat is low-grade heat sources with a temperature lower than 573 K,[1] the utilization of waste heat is usually costly and inefficient. Organic Rankine cycle (ORC) has attracted much attention due to its ability to generate electricity from low-temperature heat sources (373–573 K). However, the thermal efficiency of laboratory-scale ORC systems is usually lower than 12%,[2] and the exergy efficiency is usually lower than 50%.[3]

The selection of different working fluids affects the efficiency of the ORC system and the stability of the system. Hundreds of organic working fluids used in ORC have been studied,[4] resulting in some heuristic guidelines[5] and quantitative criteria.[6] Bao and Zhao[7] screened pure working fluids and mixed working fluids for ORC and empirically summarized the working fluids suitable for different heat source temperatures. He et al.[8] proposed an ORC thermodynamic model and summed up the selection criteria of ORC working fluid for different heat sources. Wang and Zhao[9] studied binary mixtures of fluorocarbons with different proportions and calculated their performance in solar power generation. Optimization via a mathematical model has also been conducted to screen both pure working fluids and binary mixtures for ORCs, using various algorithms, such as evolutionary

Abbreviations: ANN, artificial neural network; CAMD, computer-aided molecular design; CFC, chlorofluorocarbon; EOS, equation of state; GC, group contribution; GWP, global warming potential; HCFC, hydrochlorofluorocarbon; HEN, heat exchange network; HFC, hydrofluorocarbon; HFO, hydrofluoroolefin; MAE, mean absolute error; MRE, mean relative error; MILP, mixed integer linear programming; MINLP, mixed integer nonlinear programming; ORC, organic Rankine cycle; PHS, process hot streams; PCS, process cold streams; RMSE, root mean square error; SMILES, simplified molecular input line entry system; TAC, total annual cost.

Xiaodong Hong and Xuan Dong contribute equally.

optimization algorithms,[10] genetic algorithms (GA),[11] and particle swarm optimization (PSO).[12] Scaccabarozzi et al.[13] screened dozens of conventional pure fluids and binary mixtures as possible candidates and used the evolutionary optimization algorithm PGS-COM to determine the maximum thermal efficiency of each working fluid.

However, simply screening existing working fluids proves inadequate to unlock the potential of ORC systems. One significant challenge lies in the vastness of the molecular space. Moreover, the environmental factor should be considered. The removal of chlorofluorocarbons (CFCs) and hydrochlorofluorocarbons (HCFCs), which were widely used in the refrigeration and geothermal industries over the last century, has been executed with the Montreal Protocol.[14] The successors of CFCs and HCFCs, hydrofluorocarbons (HFCs), will also be phased out within decades[15] because of the high GWP (global warming potential). Hydrofluoroolefins (HFOs), as a new generation of low GWP working fluids, have been proposed as an alternative to HFCs. Brown et al.[16] found that hydrofluoroolefins (HFOs) are easily degraded in the troposphere due to the presence of at least one double bond. Additionally, the replacement of some hydrogen atoms in HFOs by halogen atoms reduces its flammability and increases its stability. Nair[15] believes that HFOs would become the best choice for replacing HFCs.

To further explore the potential of ORC systems, the researches have been extended to the integrated design of new working fluids and ORC systems. It aims to find new working fluids for better ORC performances. The integrated design of ORC and working fluids leads to optimization problems consisting of an ORC model and a thermodynamic property model of the working fluid. The thermodynamic property model can be achieved by the equation of state (PR EOS, SRK EOS, etc.)[17] and regression models, such as polynomial regression models.[18–21] However, the surrogate models are all modeled for a single working fluid, and there is no general surrogate model that can calculate the power output of all working fluids. To avoid the prohibitive size and complexity of the integration model of an ORC model and a thermodynamic model, targeting approaches have been developed allowing for the identification of favorable fluid properties.[22,23] Targeting approaches can also be regarded as reverse strategies. In the targeting stage, real working fluids are not identified, but properties of a working fluid that indicate a favorable performance are found. Then, a working fluid is designed to match the desired properties. Lampe et al.[24] developed the continuous-molecular targeting computer-aided molecular design (CoMT-CAMD) method, where the perturbed chain-statistical associating fluid theory (PC-SAFT) EOS was used for modeling working fluids. Four PC-SAFT parameters of a hypothetical working fluid and four process variables were identified in the first stage. Then, working fluids were obtained by a mixed integer quadratic program, where a group contribution method of PC-SAFT parameters was applied. Schilling et al.[25,26] extended the two-stage CoMT-CAMD to one-stage CoMT-CAMD. The author then applied the above method to the simultaneous design of the optimal molecule, process, and flowsheet of ORC[27] for a single waste heat source. Kleef et al.[28] proposed a SAFT-γ Mie-based CAMD-ORC optimization framework, where GCM is used to estimate critical and transport properties and detailed equipment sizing and system cost estimation models are included. However, the average error of GCM prediction is about 10%.[29]

For new working fluids, such as HFO, there is a very lack of thermodynamic data and EOS parameters in the existing literature and databases, which increases the difficulty of HFO selection for ORC. Due to the advances in artificial intelligence, many structural property relationships have been developed in various fields, such as protein structure prediction,[30] metal–organic framework property prediction,[31] and retrosynthesis prediction.[32] With a certain amount of thermodynamic property data of working fluids, the thermodynamic properties of new molecules can also be estimated by computer-aided molecular design (CAMD) methods, including the group contribution method (GCM),[33] quantitative structure–property relationship (QSPR),[34,35] and artificial neural network (ANN).[36,37] Alshehri et al.[38] compared two property estimation models, GCM and group contribution-artificial neural network (GC-ANN), utilizing functional groups for molecular representation. As compared with classical GCM, GC-ANN showed significant improvements. The thermodynamic models can be categorized into two types. The first type directly constructs the relationship between molecules and the thermodynamic models. Palma-Flores et al.[29,39] adopted the group contribution method to estimate the thermodynamic properties of new working fluids and constructed a MINLP model for the optimization of molecular groups and ORC. The second type involves an EOS parameter estimation model and an EOS function, where the EOS parameter estimation model constructs the relationship between molecules and the EOS parameters. Cignitti et al.[40] combined GCM and the SRK equation to build an optimization model to design new molecules suitable for diesel exhaust heat recovery. The integration of such thermodynamic property prediction models and the HEN-ORC model to explore new working fluids increases the difficulties of the optimization, because of the complexity of the property prediction models.

In addition, the integration design of ORC and HEN can recover the exergy losses of the HEN and improve ORC efficiency.[41] The integrated design of HEN-ORC and working fluids could be an effective way to improve the system's performance. In our previous work,[21] we developed a two-stage optimization strategy to screen working fluids in 79 pure and mixture candidates. GA was adopted in the outer layer to optimize the working fluid, and the NLP model was built in the inner layer for the optimization of ORC-HEN. To the best of our knowledge, the integrated design of HEN-ORC and new working fluid has been seldom reported. Challenges and difficulties remain for such multi-scale optimization problems, across molecules, processes, and networks. (1) Extending current one-stage and two-stage reverse methods from ORC systems to HEN-ORC systems will encounter more challenges, as HEN itself is a very difficult NP-hard problem to solve. (2) Current two-stage reveres methods use EOS parameters to integrate the first and the second stages. They are generally obtained via experiments that are considerably difficult to estimate. Besides, such two-stage methods need to include EOS constraints in the HEN-ORC model, which is generally highly nonlinear. (3) The chemical feasibility and stability of the obtained molecules from group contribution methods cannot be guaranteed when used to identify new working fluids that match desired properties.

A two-stage reverse strategy using an EOS-free HEN-ORC model is developed to overcome the contradiction. Specifically, a quantitative heat-to-work model of ORC is developed, enabling the calculation of the power output based on five thermodynamic properties of the working fluid, including heat capacities and vaporization enthalpies. Then, with these five thermodynamic properties, process operating conditions, and network structures as decision variables, the first-stage optimization aims to determine the optimal ORC process and HEN structure, as well as the thermodynamic properties that the optimal fluid should possess. This approach avoids thermodynamic calculations involving EOS, significantly reducing the complexity of integrated design of ORC-HEN and working fluids. In the second stage, the best HFO working fluids are directly identified from a database containing more than 430,000 hydrofluoroolefins (HFOs), by minimizing the error between the predicted and optimal thermodynamic properties. Two group contribution-artificial neural network prediction models are developed to predict the thermodynamic properties of HFOs in our database.

2 | PROBLEM DESCRIPTION

The ORC working fluid selection and ORC-HEN process design problems are described as follows: Given the stream data for a certain process, including a set of process hot streams PHS = {i|1, …, I} and a set of process cold streams PCS = {j|1, …, J}. The inlet temperatures (TIN_i, TIN_j), outlet temperatures ($TOUT_i$, $TOUT_j$), and heat capacity flow rates (FCp_i, FCp_j) of both hot and cold process streams are provided. It is required to determine the optimal ORC working fluid, ORC operating parameters, and HEN-ORC structure, which gives the lowest total annual cost. Working fluids are screened in the HFO molecule library (C2–C10, from the GDB molecule generation library, a total of 430,000 molecules). The following assumptions are adopted:

a. The heat capacities of the hot and cold process streams are constant.
b. The heat capacities of the ORC working fluid are constant in the preheating process, the superheating process, and the desuperheating process.
c. The inlet and outlet temperatures of the pump are equal since the temperature rise of pumping is generally small and negligible.

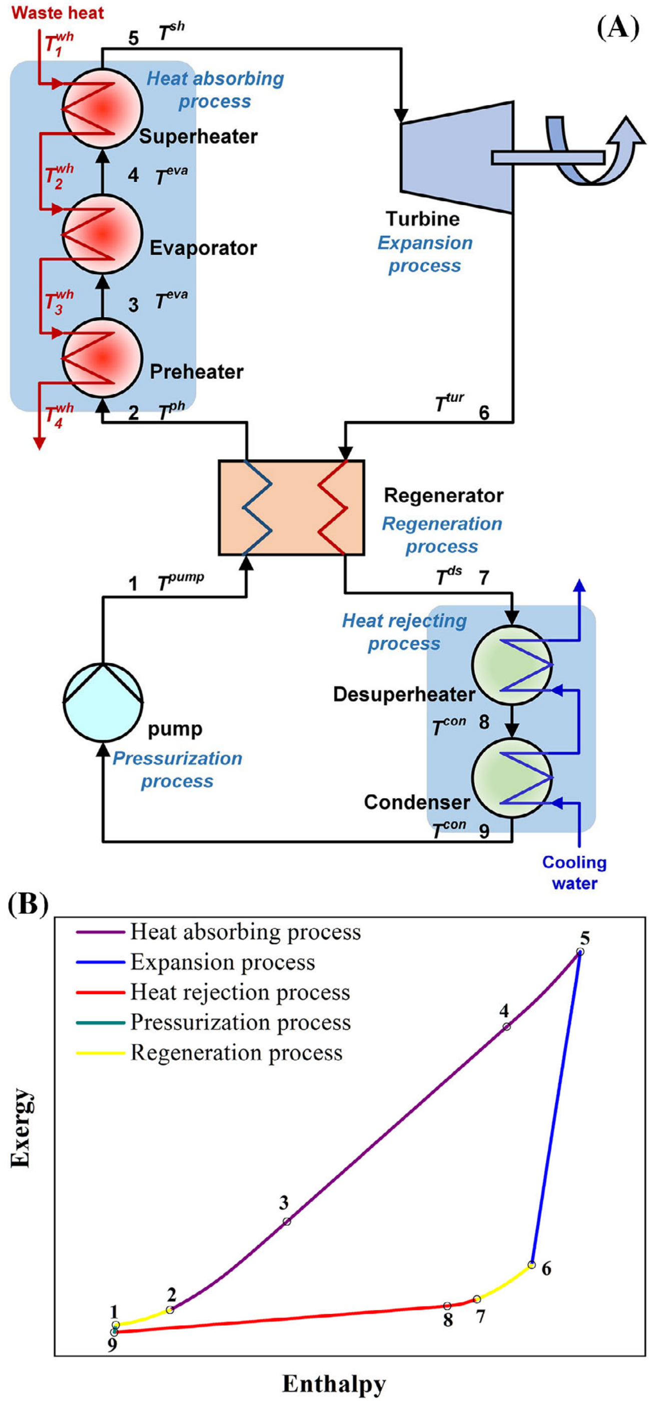


FIGURE 1 (A). The flowsheet of ORC. (B) ORC exergy-enthalpy diagram.

3 | ORC EXERGY ANALYSIS AND HEAT-TO-WORK MODEL

ORC is mainly composed of basic processes such as heat-absorbing process, expansion process, heat rejection process, and pressurization process. In addition, the regeneration process is often used to improve ORC efficiency in the industry. Figure 1A shows the flowsheet of ORC, in which the heat-absorbing process includes a preheater, an evaporator, and a superheater. The expansion process is implemented by the turbine, the regeneration process includes a regenerator, and the heat rejection process includes a desuperheater and condenser. The pressurization process is completed by the pump. The regeneration process is used for self-heat exchange between the expanded high-temperature gas and the compressed low-temperature liquid. That is, the ORC system can be divided into two parts: the heat exchange process and the work process. The heat exchange process needs to meet the constraints of the temperature difference and heat load.

To calculate the power output, which is a significant index to evaluate the performance of ORC, complex EOS methods are generally applied, such as PC-SACT.[24] In this study, we build an EOS-free heat-to-work model of ORC, where the power output is only calculated from the heat capacities and vaporization enthalpies of the working fluid. This is achieved through exergy analysis, which can intuitively represent the maximum work potential of the heat source. The change of exergy and enthalpy in ORC is described in the exergy-enthalpy diagram, as shown in Figure 1B. The heat-to-work model is derived from the first principle thermodynamic equations. The power output can be obtained via the following equation:

$$W_{net} = \eta_{ise}\left(\left(1-\frac{\overline{T}^{rej}}{\overline{T}^{abs}}\right)\Delta H^{abs} - \frac{\overline{T}^{rej}}{T_0}\Delta Ex_{loss}^{reg}\right) \tag{1}$$

where W_{net} of the ORC system depends on $\overline{T}^{abs}$, $\overline{T}^{rej}$, ΔH^{abs}, and ΔEx_{loss}^{reg}, which are determined by five ORC operating temperatures

$$\Phi = \frac{\left|F^{wf}Cp^{ph} - \tilde{F}^{wf}\tilde{Cp}^{ph}\right|\left(T^{eva}-T^{ph}\right) + \left|F^{wf}\Delta h^{eva} - \tilde{F}^{wf}\tilde{\Delta h}^{eva}\right| + \left|F^{wf}Cp^{sh} - \tilde{F}^{wf}\tilde{Cp}^{sh}\right|\left(T^{sh}-T^{eva}\right)}{\Delta H^{abs}} + \frac{\left|F^{wf}\Delta h^{con} - \tilde{F}^{wf}\tilde{\Delta h}^{con}\right| + \left|F^{wf}Cp^{ds} - \tilde{F}^{wf}\tilde{Cp}^{ds}\right|\left(T^{ds}-T^{con}\right)}{\Delta H^{rej}} \tag{2}$$

(T^{ph}, T^{eva}, T^{sh}, T^{con}, and T^{ds}) and five thermodynamic properties of working fluids (Cp^{ph}, Cp^{sh}, Cp^{ds}, Δh^{eva} and Δh^{con}). $\overline{T}^{abs}$ and $\overline{T}^{rej}$ are defined as the mean temperature of the heat-absorbing and heat rejection processes. ΔH^{abs} is the enthalpy change of the heat-absorbing process, and ΔEx_{loss}^{reg} is the exergy loss between the hot and cold sides of the regenerator. The detailed introduction of the ORC exergy analysis and the heat-to-work model is presented in the Data S1.

4 | REVERSE STRATEGY AND EOS-FREE HEN-ORC MODEL

The proposed method aims to model the HEN-ORC with a hypothetical working fluid without EOS, where the power output of the ORC can be directly calculated from the thermodynamic properties and operating temperatures of the heat process. Thus, the ideal HEN-ORC with unknown working fluids can be optimized using the HEN-ORC optimization model. The reverse strategy for the integrated design of HEN-ORC and HFO working fluids is summarized as follows, and its schematic illustration is shown in Figure 2. In the first stage, we obtain stream matching results of HEN, the optimal temperature parameters (T^{ph}, T^{eva}, T^{sh}, T^{con}, and T^{ds}), and thermodynamic properties (Cp^{ph}, Cp^{sh}, Cp^{ds}, Δh^{eva}, and Δh^{con}) of the ideal ORC for the specific background streams, by solving the EOS-free HEN-ORC model. Then, before carrying out the second stage, two working fluid databases are constructed. Database #1 comprises 430,000 molecules of HFO, serving as potential working fluids. Database #2 is built to provide physical property data and molecular groups for the GC-ANN model. The trained GC-ANN model is used to predict T^c, Cp^{ph}, Cp^{sh}, Cp^{ds}, Δh^{eva}, and Δh^{con} of the HFO in Database #1.

In the second stage, the HFOs are filtered from Database #1, whose predicted values of thermodynamic properties are similar to the optimized values. This process includes the following steps:

i. Ensure HFOs operating below the critical state, HFOs whose T^c is less than T^{eva} are eliminated.
ii. Calculate the error (Φ) between the predicted value and the optimized value of Cp^{ph}, Cp^{sh}, Cp^{ds}, Δh^{eva}, and Δh^{con}, and sort the remaining HFOs according to Φ. Φ is calculated by the following equation:

where $\tilde{Cp}^{sh}$, $\tilde{Cp}^{ds}$, $\tilde{Cp}^{ph}$, $\tilde{\Delta h}^{eva}$, and $\tilde{\Delta h}^{con}$ are the predicted values of the thermodynamic properties, and $\tilde{F}^{wf}$ is the predicted flow rate and expressed by the following equation:

$$\tilde{F}^{wf} = \frac{\Delta H^{abs}}{\widetilde{Cp}^{ph}\left(T^{eva}-T^{ph}\right) + \widetilde{\Delta h}^{eva} + \widetilde{Cp}^{sh}\left(T^{sh}-T^{eva}\right)} \tag{3}$$

iii. Find the HFO with the smallest Φ as the selected working fluid, and re-simulate and fine tune the HEN-ORC system.

The HEN-ORC optimization model includes three parts: (a) ORC heat-to-work model; (b) expanded stage-wise superstructure involving ORC; and (c) Constraints on thermodynamic properties of actual working fluids. The ORC heat-to-work model from Section 2 is used to calculate the heat and work process. The expanded stage-wise superstructure involving ORC from our previous work[21] is adopted and revised to optimize heat matches of all streams by adding the ORC superheating process and regeneration process. The detailed equations are presented in Data S1. Besides, since no specific working fluid is introduced during the optimization of the ideal HEN-ORC, constraints on the thermodynamic properties of working fluids are deployed to avoid impractical results. Thermodynamic properties as variables are constrained in this work. The bounds of practical working fluids are described by Equations (4–9). Equations (4–7) are

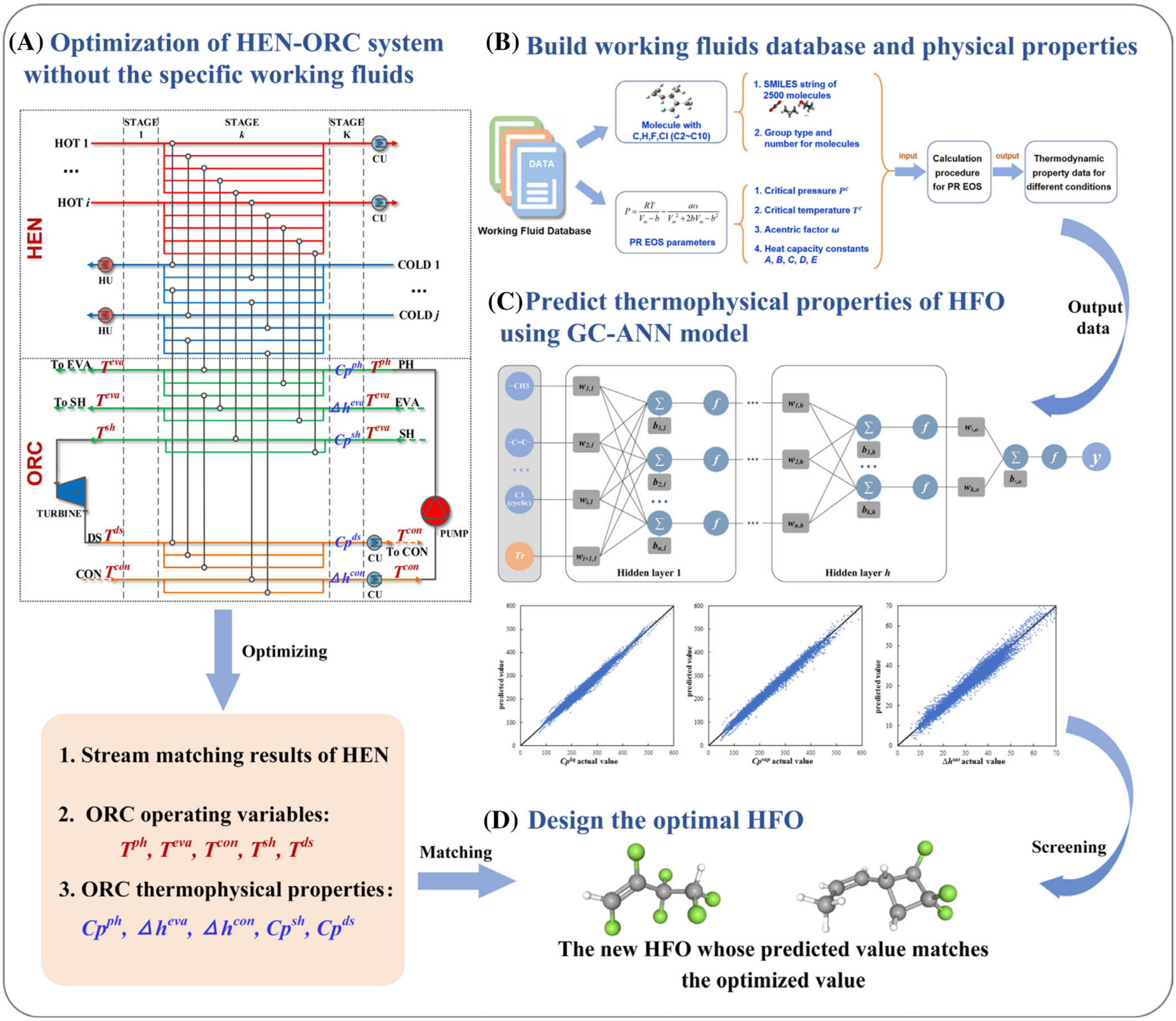


FIGURE 2 The schematic illustration of the integrated design of HEN-ORC and HFO working fluids.

regressed from all molecules of Database #2, as shown by Figure S3 of the Data S1.

$$20 \le \frac{\Delta h^{eva}}{Cp^{ph}} \le 302.8 - 0.306T^{eva} \tag{4}$$

$$1.097 - 1.237 \times 10^{-3}T^{eva} \le \frac{Cp^{sh}}{Cp^{ph}} \le 1.637 - 1.464 \times 10^{-3}T^{eva} \tag{5}$$

$$0.941 - 8.734 \times 10^{-4}T^{eva} \le \frac{\Delta h^{con}}{\Delta h^{eva}} \le 2.291 + 9.113 \times 10^{-3}T^{eva} \tag{6}$$

$$0.4 \le \frac{Cp^{ds}}{Cp^{ph}} \le 0.953 - 6.848 \times 10^{-4}T^{eva} \tag{7}$$

$$60 \le T^{eva} \le 300 \tag{8}$$

$$0.9P^{c} \le T^{eva} \tag{9}$$

The model for the HEN-ORC optimization can be formulated as the following problem P1.

$$\text{Min } TAC \text{ or } W_{net}$$

$$st.\left[\begin{array}{l} Equation\,(1)\ and\ Equations\,(S6-S29)\ of\ Supporting\ Information \\ Equations\,(S32-S98)\ of\ Supporting\ Information \\ Equations\,(4-9) \end{array}\right. \tag{P1}$$

where Equation (1) and Equations (S6-S29) of the Data S1 represent the ORC heat-to-work model, Equations (S32-S98) of the Data S1 represent the expanded stage-wise superstructure involving ORC, and

Equations (4–9) are the constraints on thermodynamic properties of working fluids.

5 | WORKING FLUID DATABASE #1 AND #2

Hydrochlorofluorocarbons (HCFCs) and hydrofluorocarbons (HFCs) commonly used in ORCs have high global warming potential (GWP), while efficient and environmentally friendly working fluids are the trend of current ORCs. Further restrictions on greenhouse gas emissions under the Kyoto Protocol, HCFCs, and HFCs will be phased out over the next few decades. Hydrofluoroolefins (HFOs) are considered fourth-generation working fluids because of their short atmospheric lifetime, resulting in extremely low GWP values. However, the lack of data on the thermodynamic properties of HFOs hinders the development of new HFO working fluids. To the best of our knowledge, there are only 23 kinds of HFOs in physical property databases such as NIST and DIPPR, as shown in Table 1, and they mainly belong to C2–C4. These working fluids cannot fully satisfy the needs of waste heat recovery in a large temperature range. To find more potential HFOs working fluid for ORCs, about 430,000 low-carbon HFOs (C2–C10) are identified and screened from the GDB molecule library, which is generated using C, N, O, and F atoms under consideration of simple valence, chemical stability, and synthetic feasibility rules.[42] The established HFO database (namely Database #1) is detailed in the Data S2.

TABLE 1 HFOs with existing physical property data.

HFOs (SMILES form)
C=C(F)F, C(=C/F)\F, C(=C\F)\F, C(=C(F)F)F, C=CF, CC=CF, C(=C(F)F)C(F)(F)F, C=CC(F)(F)F, C1 = CC1(F)F, C=CCF, C(=C\F)\C(F)(F)F, C(=C(/C(F)(F)F)\F)\F, C=C(C(F)(F)F)F, CC=C(F)F, C(=C/F)\C(F)(F)F, C=CC(=C)F, CC(=C)C(F)(F)F, C1CC1 = C(F)F, C=C1CC1(F)F, C1CC(CC=C1)F, C=CC1 = CC=CC=C1F, C=CC1 = CC(=CC=C1)F, C=CC1 = CC(=CC=C1)C(F)(F)F

As mentioned above, five thermodynamic data are needed to calculate the p output of ORC. They are Cp^{sh}, Cp^{ds}, Cp^{ph}, Δh^{eva}, and Δh^{con}. In the absence of physical property data for HFOs, it is significant to accurately predict the physical property data of undeveloped HFOs. The group contribution method gives some enlightenment, and the combination of groups shows a good prediction of thermodynamic properties and transport properties. Although the physical properties of most HFOs are unknown, the groups making up HFOs can be found in other hydrocarbons with known data. Therefore, a hydrocarbon working fluid database (namely Database #2 shown in the Data S2) is constructed in this work to predict the physical properties of HFOs. To increase the amount of working fluid data, the database not only includes hydrocarbons containing C, H, and F elements, but also includes hydrocarbons containing C, H, F, and Cl elements. These hydrocarbons also contain C=C and other groups constituting HFOs. The working fluid data are from NIST and DIPPR.

Thermodynamic properties under different conditions for hydrocarbons in Database #2 can be obtained, as shown in Figure 3. The working fluid database includes more than 2500 kinds of C2–C10 chain hydrocarbons, cyclic hydrocarbons, and aromatic hydrocarbon molecules. Each molecule has its PR equation parameters, including critical temperature T^c, critical pressure P^c, eccentricity factor ω, and Aly and Lee gas heat capacity constants (A, B, C, D, E),[43] where the equation for the ideal gas heat capacity of Aly and Lee is as follows:

$$c_p(T) = A + B\left(\frac{\frac{C}{T}}{sinh\left(\frac{C}{T}\right)}\right)^2 + D\left(\frac{\frac{E}{T}}{cosh\left(\frac{E}{T}\right)}\right)^2 \tag{10}$$

6 | GC-ANN MODEL

Not only the above-mentioned five thermodynamic properties (Cp^{sh}, Cp^{ds}, Cp^{ph}, Δh^{eva}, and Δh^{con}), but also the critical temperatures (T^c) of working fluids, should be predicted in this article. This is because we only study ORCs in the subcritical state, which requires T^c be greater

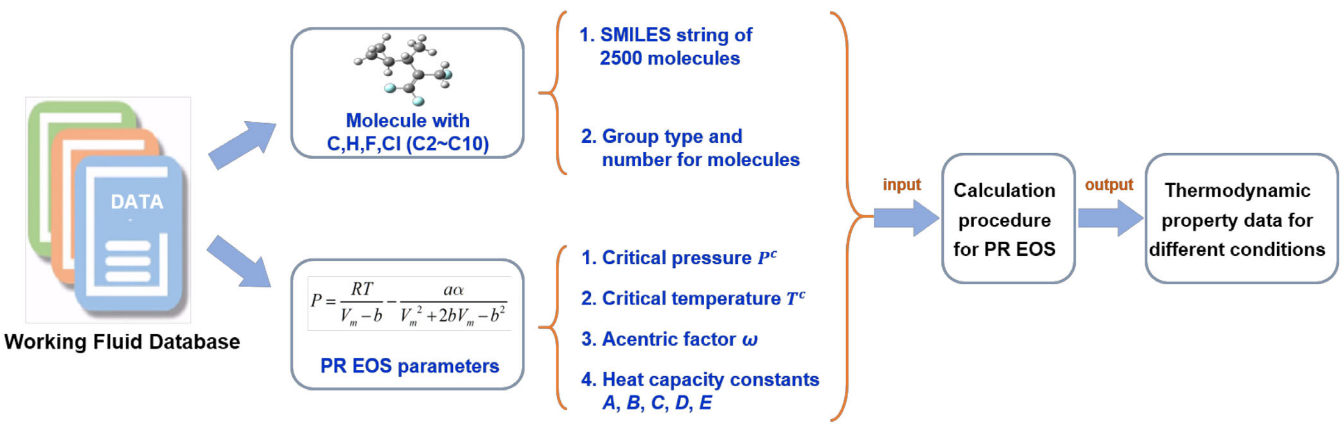


FIGURE 3 Hydrocarbon working fluid Database #2 and the calculation of thermodynamic property data. The methodology to calculate thermodynamic property is introduced in the Data S1.

than T^{eva}. T^c is the inherent property of the working fluid, while the five thermodynamic properties are not only related to the working fluid, but also depend on T^{sat}. Cp^{sh} and Cp^{ds} are both vapor heat capacities at different operating temperatures, Cp^{ph} is liquid heat capacity, and Δh^{eva} and Δh^{con} are saturated vaporization enthalpies at different operating temperatures. Thus, these five thermodynamic properties can be further classified as three physical parameters for prediction: vapor heat capacity (Cp^{vap}), liquid heat capacity (Cp^{liq}), and saturated vaporization enthalpy (Δh^{sat}). Consequently, we will build two GC-ANN models (ANN1 and ANN2), to predict T^c and the other five thermodynamic properties, respectively. First-order and second-order groups of molecules in Database #2 are identified, as shown in Table S1 of the Data S1.

ANN1 is used to train T^c, with the type and number of molecular groups as input and T^c as output. The molecular groups and T^c data are from Database #2. ANN2 is used to train the other five parameters. To prevent T^{eva} greater than T^c of the working fluid, the relative saturation temperature T_r ($T_r = \frac{T^{sat}}{T^c}, 0 < T_r < 1$) is introduced to replace T^{sat} in ANN2. Thus, ANN2 takes the type and number of molecular groups and T_r as input, and Cp^{vap}, Cp^{liq}, and Δh^{sat} as output. The data are calculated based on Database #2.

The GC-ANN model is developed using TensorFlow 2.0. Figure 4 shows the training framework of the developed model. First, the molecules of Database #2 are randomly divided into three groups: training set (70%), validation set (10%), and test set (20%). The training set, validation set, and test set contain 1757, 251, and 503 molecules, respectively. The above ANN1 and ANN2 are trained by the training set, respectively. Figure 4 also shows the neural network structure, in which the network consists of h hidden layers, x represents the input, y indicates the output, and w and b are the weights and biases of the neurons. The initial values of weights and biases are randomly assigned. The transfer function adopts the RELU function. The optimization algorithm (Adam) is used to optimize the weights and biases. The mean square error (MSE) is used as the objective function.

ANN1 uses 3 hidden layers to predict T^c. The number of neurons in each layer is [8, 16]. Figure 5A shows the T^c predicted and actual values of the test set. Four statistical metrics, root mean square error (RMSE), mean absolute error (MAE), mean relative error (MRE), and correlation coefficient (R^2), are used to evaluate the accuracy of the model in predicting T^c, listed in Table 2. RMSE, MAE, MRE, and R^2 of the test set are 19.48%, 13.9%, 2.40%, and 0.948. The results are better than that of Su et al.[35] where the MAE of the training set and MAE of the test set are 22.48 and 22.77, and the MRE of the training set and MRE of the test set are 4.23% and 5.29%, respectively. The prediction error of the ANN1 model is relatively small, which demonstrates the accuracy of the model in predicting the T^c of hydrocarbons.

ANN2 also uses 3 hidden layers to predict Cp^{liq}, Cp^{vap}, and Δh^{sat}, and the number of neurons in each layer is [64, 64, 32]. T_r is valued between 0 and 1 with a step size of 0.03, and the data whose saturation temperature is lower than 273.15 K are deleted from the training group. The number of data in the training set, validation set, and test set are 31,005, 4285, and 8854, respectively. Figure 5B–D shows the predicted and actual values of Cp^{liq}, Cp^{vap}, and Δh^{sat} in the test set, respectively. The same statistical metrics as ANN1 are used to evaluate the prediction accuracy, as shown in Table 3. The prediction accuracy of Δh^{sat} is similar to that of Yan et al.,[34] where R^2 and MAE of the test set are 0.967 and 2.499. In addition, Alshehri et al.[38] predicted the Δh^{sat} of organic molecules using machine learning, where

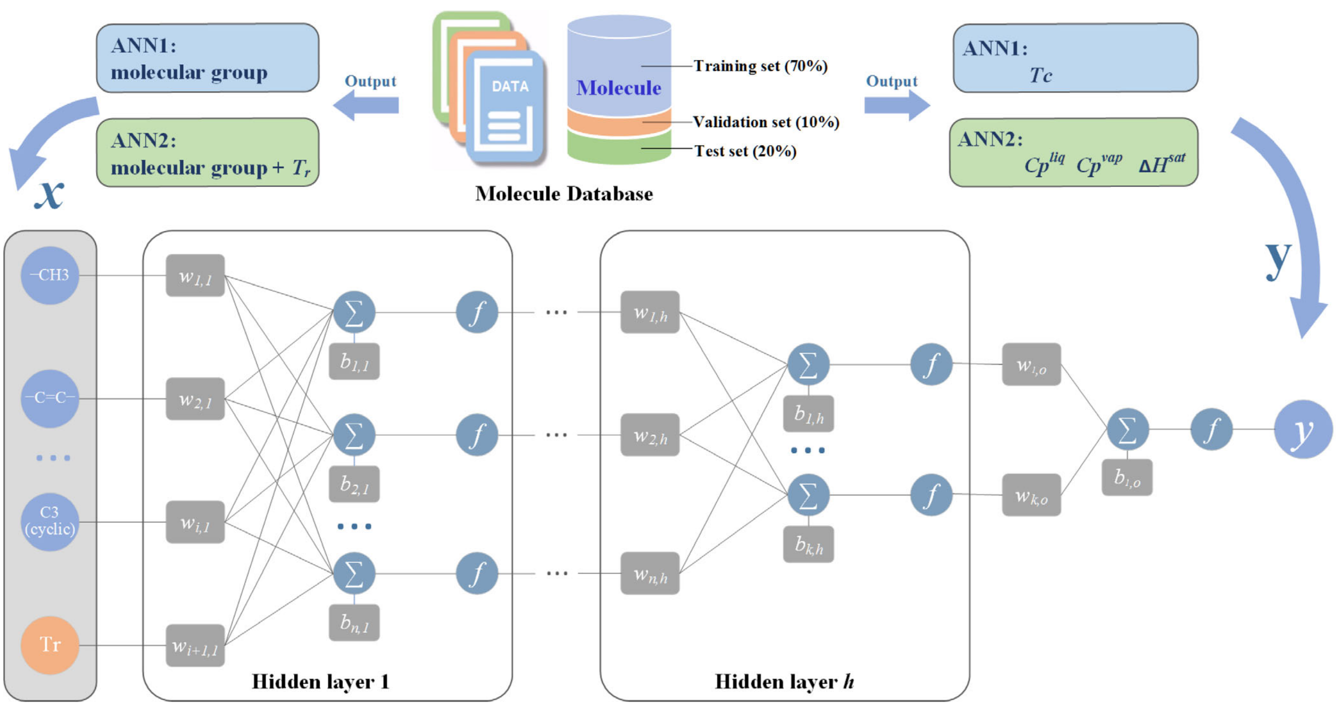


FIGURE 4 GC-ANN model framework.

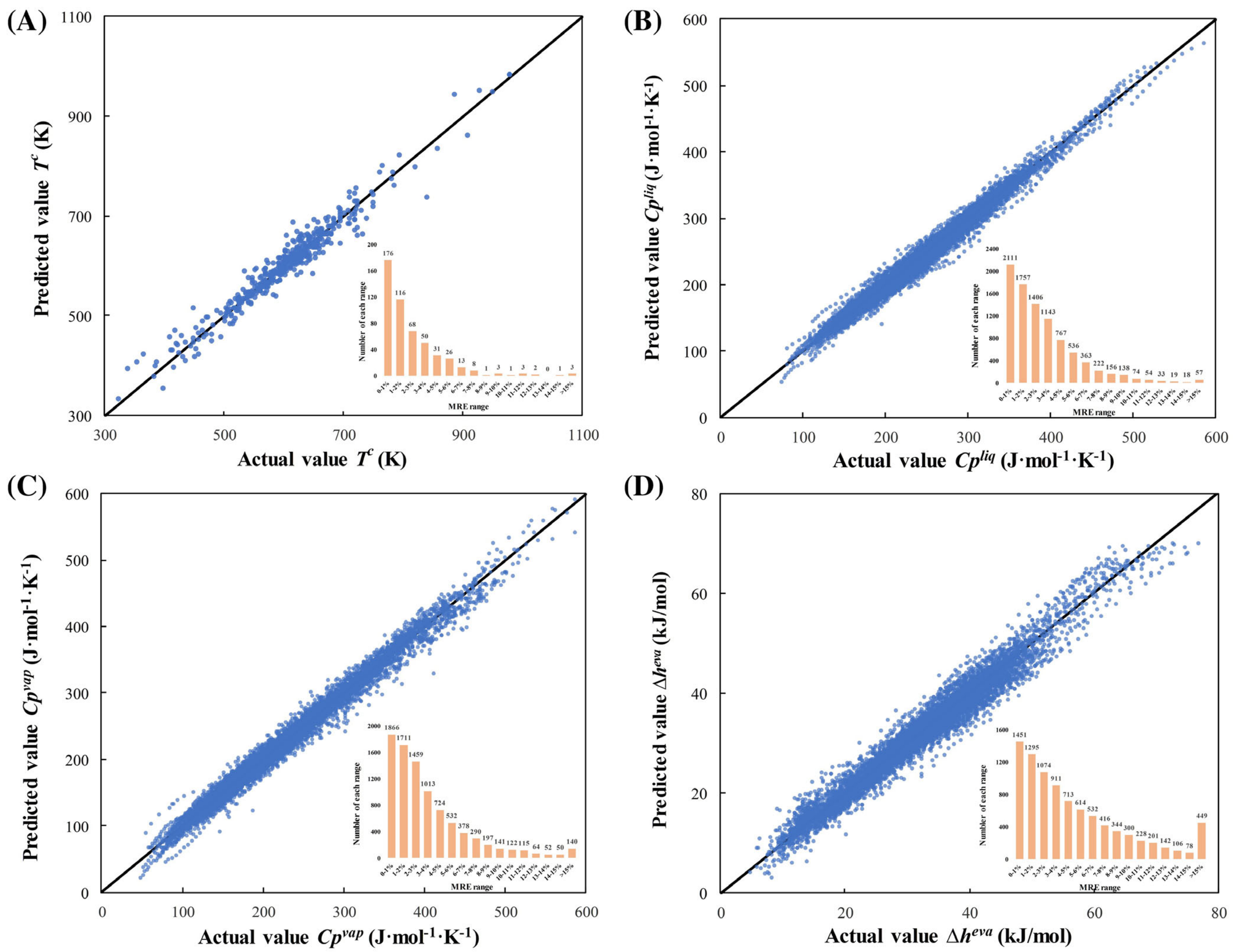


FIGURE 5 Comparison of the predicted and actual values and MRE distribution for the test set. (A) T^c, (B) Cp^{liq}, (C) Cp^{vap}, (D) Δh^{sat}.

TABLE 2 Statistical results of the training set, validation set, and test set in ANN1.

T^c	Train set	Validation set	Test set
RMSE	15.36	15.75	19.48
MAE	10.1	10.5	13.9
MRE	1.76%	1.84%	2.40%
R^2	0.972	0.977	0.948

the R^2 and RMSE of the test set are 0.965 and 2.47. Although the organic molecules used for training in the above literature are not the same as those in our working fluid database, the similar prediction levels indicate that the model has a good predictive ability.

7 | CASE STUDY

The proposed method is applied to two literature cases from our previous work[21] and Desai and Bandyopadhyay,[44] respectively.

7.1 | Case 1

In the previous work,[21] we designed HEN and ORC simultaneously with n-pentane as the working fluid by minimizing TAC. The stream data of two hot streams, two cold streams, hot utility, and cold utility are given in Table 4, as well as the heat transfer coefficient of ORC streams with different phases. The economic parameters are shown in Table 5. The minimum temperature difference is assumed as 10 K.

Following the procedures of Figure 2, P1 is used to optimize the heat transfer matching of HEN, the ORC operating parameters, and the thermodynamic properties of the ideal working fluid. The optimization results are shown in Table 6. The obtained structure of HEN-ORC is shown in Figure 6, in which SH, EVA, PH, and CON indicate superheater, evaporator, preheater, and condenser, respectively. No regenerator and desuperheater are required by the obtained HEN-ORC since the turbine expands to the condensation temperature. The net power output of the ideal ORC is 5760.8 kW, and the TAC of the ideal HEN-ORC is as low as 2.565 M$/year.

TABLE 3 Statistical results of the training set, validation set, and test set in ANN2.

Parameters		Train set	Validation set	Test set
Cp^{liq} /J·mol^{-1}·K^{-1}	RMSE	7.88	10.45	9.37
	MAE	5.68	7.48	7.19
	MRE	2.47%	3.24%	3.11%
	R^2	0.988	0.980	0.981
Cp^{vap} /J·mol^{-1}·K^{-1}	RMSE	9.15	11.45	10.58
	MAE	6.28	7.82	7.80
	MRE	2.90%	3.63%	3.66%
	R^2	0.989	0.984	0.985
Δh^{sat} /kJ·mol^{-1}	RMSE	1.87	2.09	2.07
	MAE	1.22	1.50	1.53
	MRE	4.27%	5.29%	5.31%
	R^2	0.970	0.961	0.964

TABLE 4 Stream data of Case 1.

Streams	T_{in} (K)	T_{out} (K)	FCp (kW/K)	h (kW/m^2·K)
H1	460.15	350.15	300	1
H2	400.15	300.15	500	1
C1	420.15	490.15	600	1
C2	320.15	390.15	200	1
HU	573.15	523.15		5
CU	288.15	303.15		1
ORC heat transfer coefficient				
Liquid streams				1
Evaporating streams				3.6
Vapor streams				0.06
Condensing streams				1.6

According to the optimized ORC operating temperatures and thermodynamic properties in Table 6, a new HFO is selected from 430,000 HFO molecules in Database #1. Firstly, the ANN1 model is used to predict the T^c of HFOs, and the prediction result shows T^c of more than 2600 low-carbon HFOs is lower than T^{eva} (390.2 K). The remaining HFOs with T^c greater than T^{eva} are screened. Then, the thermodynamic properties at the operating temperatures of all screened HFOs are predicted by the ANN2 model. The error Φ between the predicted values and the optimal values of each HFO is calculated by Equation (1), and HFOs are sorted to find the optimal HFO working fluid with the smallest Φ. Table 7 shows the 5 new HFOs with the smallest error and their predicted thermodynamic properties. The optimal HFO (FC1 = CC2C3CC2(F)CC13F) with the smallest error of 0.072 is selected as the ORC working fluid.

The HEN-ORC is obtained without a specific working fluid, and the thermodynamic properties of the selected working fluid might not satisfy the obtained HEN-ORC. The obtained system must be re-simulated using FC1 = CC2C3CC2(F)CC13F as the working fluid.

TABLE 5 Economic parameters for Case 1.

Parameters	Values	Parameters	Values
CCR (year^{-1})	0.2	S_{REF}^{pump} (kW)	1000
MF	2.5	α_{tur}	0.67
C_F^{ex} (\$)	100	α_{pump}	0.67
C_A (\$/m^2)	400	η_{ise}^{tur}	0.8
A_{REF} (m^2)	500	η_{ise}^{pump}	0.65
β_1	0.6	C_{CU} (\$/kW·year)	20
β_2	1	C_{HU} (\$/kW·year)	100
C_S^{tur} (\$/kW)	430	h_{op} (h/year)	7000
C_S^{pump} (\$/kW)	100	p_{el} (\$/kW·h)	0.14
S_{REF}^{tur} (kW)	4000		

TABLE 6 Optimization results of Case 1.

	Results	Value
ORC	T^{ph} (K)	313.15
	T^{eva} (K)	390.15
	T^{sh} (K)	400.15
	T^{con} (K)	313.15
	T^{ds} (K)	313.15
	$F^{wf}Cp^{ph}$ (kW/K)	442.6
	$F^{wf}Cp^{sh}$ (kW/K)	421.4
	$F^{wf}Cp^{ds}$ (kW/K)	/
	$F^{wf}\Delta h^{eva}$ (kW)	13,785
	$F^{wf}\Delta h^{con}$ (kW)	46,275
	W_{net} (kW)	5760.8
HEN	Q^{hu} (kW)	42,000
	Q^{cu} (kW)	63,239
	A_{total} (m^2)	15,110
	$COST_{op}$ (M\$/year)	5.465
	$COST_{cap}$ (M\$/year)	2.746
	$SALE_{el}$ (M\$/year)	−5.646
	TAC (M\$/year)	2.565

The heat exchange load, temperature, and power output of the HEN-ORC are fine-tuned and the modified HEN-ORC is shown in Figure 7. Compared to Figure 6, the parameters are fine-tuned. For example, the heat load of SH is modified from 4.21 MW to 4.48 MW, and the corresponding SH outlet temperature of the H1 stream is corrected to 445.3 K. The results of the utility load, operating cost, equipment cost, power revenue, and TAC of the modified HEN-ORC system are listed in Table 8. The TAC of the new design is 2.763 M\$/year, which is 26%, 12%, and 22% smaller than the results by Dong et al.,[21] Xu et al.,[20] and Watanapanich et al.,[18] respectively. The net power output of the obtained result is 69%, 44%, and 11% higher than the ones in the literature, respectively.

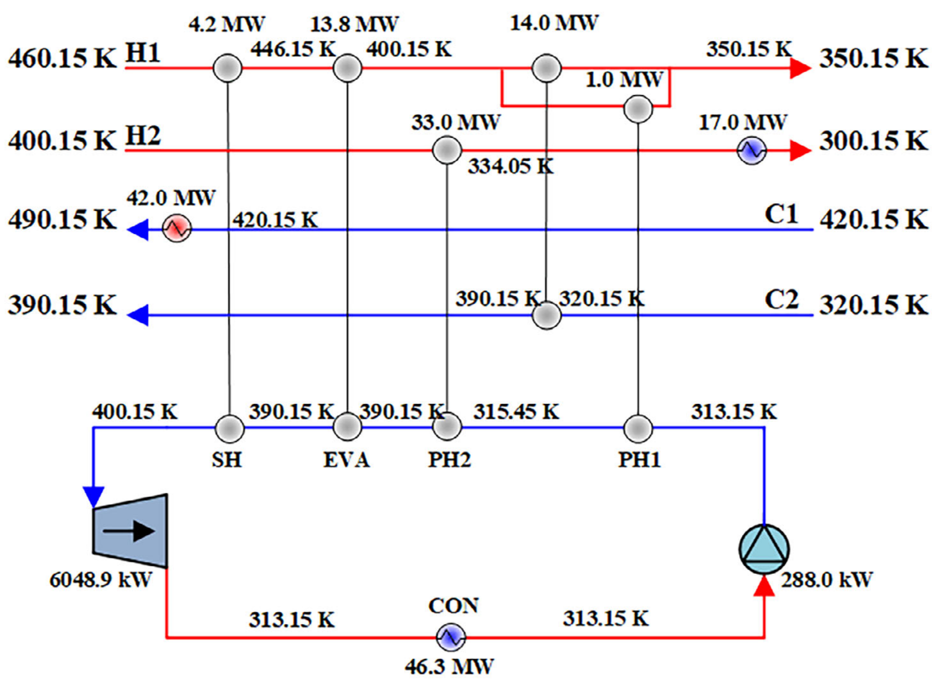
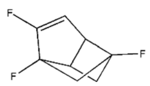
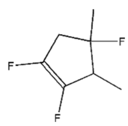
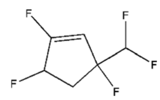
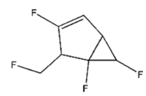
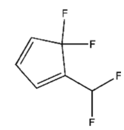


FIGURE 6 The obtained HEN-ORC results of Case 1.

TABLE 7 The 5 HFOs with the smallest error for Case 1.

Optimal HFOs	1	2	3	4	5
Molecule					
SMILES	FC1 = CC2C3CC2(F)CC13F	CC1C(F) = C(F)CC1(C)F	FC(F)C1(F)CC(F)C(F) = C1	FCC1C(F) = CC2C(F)C12F	FC(F)C1 = CC=CC1(F)F
T^c (K)	404.75	396.75	397.05	414.65	413.35
$\tilde{F}^{wf}$ (kmol/s)	1.571	2.114	1.831	2.049	2.057
$\tilde{Cp}^{ph}$ (kJ/kmol·K)	272.7	197.1	222.3	220.8	218.0
$\tilde{Cp}^{sh}$ (kJ/kmol·K)	285.0	203.3	241.3	213.8	260.5
$\tilde{Cp}^{ds}$ (kJ/kmol·K)	190.6	113.2	151.3	134.9	194.0
$\tilde{\Delta h}^{eva}$ (kJ/kmol)	8609.8	6481.6	7415.3	6262.5	5913.1
$\tilde{\Delta h}^{con}$ (kJ/kmol)	28250.1	21105.7	24764.8	23860.6	23191.2
$\tilde{F}^{wf}\tilde{Cp}^{ph}$ (kW/K)	428.3	416.7	407.1	452.3	448.3
$\tilde{F}^{wf}\tilde{Cp}^{sh}$ (kW/K)	447.7	429.8	441.9	438.1	535.7
$\tilde{F}^{wf}\tilde{Cp}^{ds}$ (kW/K)	299.4	239.2	277.1	276.4	399.0
$\tilde{F}^{wf}\tilde{\Delta h}^{eva}$ (kW)	13523.0	13702.2	13581.0	12830.4	12161.2
$\tilde{F}^{wf}\tilde{\Delta h}^{con}$ (kW)	44371.1	44617.7	45356.4	48884.7	47696.5
Φ	0.072	0.077	0.080	0.092	0.092

7.2 | Case 2

Desai and Bandyopadhyay,[44] Chen et al.,[45] and Watanapanich et al.[18] designed HEN and ORC simultaneously with the maximum W_{net} as the objective function. The stream data of three hot streams and four cold streams in this case are given in Table 9. The minimum temperature difference is 20 K.

The same design strategy of Figure 4 is applied in this case. The net power output is optimized by model P1, obtaining the heat transfer matching, the optimal ORC operating parameters, and

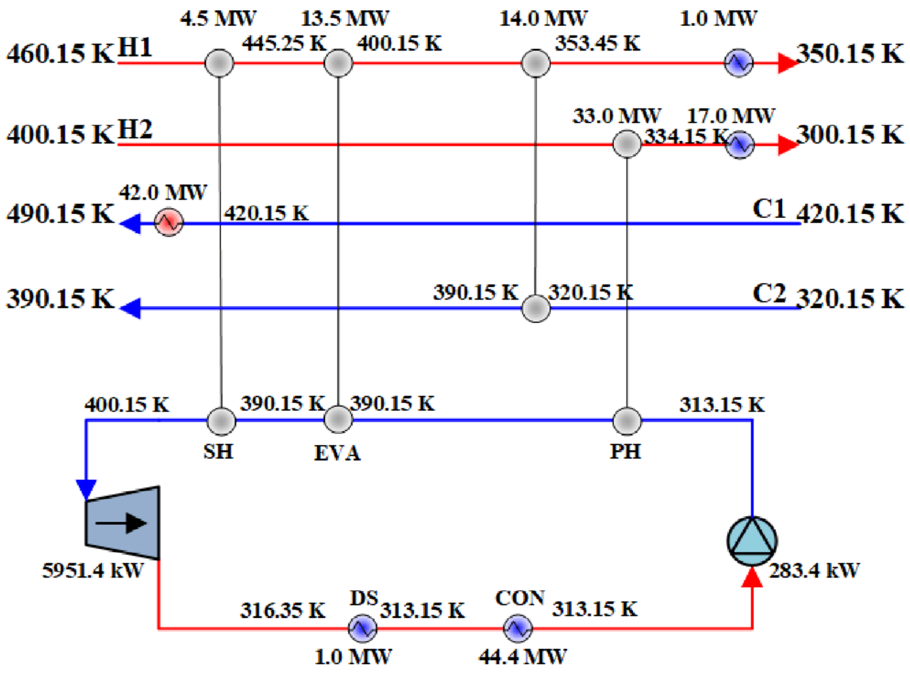
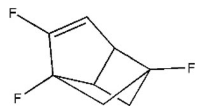


FIGURE 7 The modified HEN-ORC with new HFO as the working fluid for Case 1.

TABLE 8 The result comparisons with the literature for Case 1.

Results	This work	Dong et al.[21]	Xu et al.[20]	Watanapanich et al.[18]
Working fluids		n-Pentane	R1234ze	n-Pentane
T^{ph} (K)	313.15	313.15	315.15	313.45
T^{eva} (K)	390.15	355.35	370.15	370.95
T^{sh} (K)	400.15	/	/	/
T^{con} (K)	313.15	313.15	313.15	313.15
T^{ds} (K)	316.35	330.85	313.15	337.75
W_{net} (kW)	5668.0	3350.6	3930.9	5098.8
Q^{hu} (kW)	42,000	33,000	33,000	45,831.6
Q^{cu} (kW)	63,340	56,909	56,069	67,733
A_{total} (m^2)	15,506	12,919.9	/	/
$COST_{op}$ (M\$/year)	5.467	4.438	4.421	5.938
$COST_{cap}$ (M\$/year)	2.851	2.569	2.582	2.610
$SALE_{el}$ (M\$/year)	−5.555	−3.284	−3.852	−4.997
TAC (M\$/year)	2.763	3.723	3.151	3.551

thermodynamic properties of the HEN-ORC. The optimization results of this case are shown in Table 10. The HEN-ORC system is shown in Figure 8, in which REG and DS indicate regenerator and desuperheater respectively. The W_{net} of the obtained ORC is 47.2 kW.

Similar to procedures introduced in the last case, 410,000 HFOs are first screened by using the constraint of T^c greater than T^{eva}. Then, the thermodynamic properties at the operating temperatures of the 410,000 HFO candidates are predicted by the GC-ANN model, and the error Φ between the predicted values and the optimal values of each HFO is calculated. Table 11 shows the 5 new HFOs with the

TABLE 9 Stream data in Case 2.

Streams	T_{in} (K)	T_{out} (K)	*FCp* (kW/K)
H1	626.15	586.15	9.802
H2	620.15	519.15	2.931
H3	528.15	353.15	6.161
C1	497.15	613.15	7.179
C2	389.15	576.15	0.641
C3	326.15	386.15	7.627
C4	313.15	566.15	1.690

smallest error and their predicted thermodynamic properties. The optimal HFO (CC(=C)C1(F)CC1C(F)F) with the smallest error of 0.113 is selected as the ORC working fluid.

Finally, the ideal HEN-ORC system is corrected using the thermodynamic properties of this new HFO, and the modified HEN-ORC structure with CC(=C)C1(F)CC1C(F)F as working fluid is shown in Figure 9. Due to the differences between the predicted and optimal values of the thermodynamic properties, the heat loads of SH, EVA, REG, and CON3 are modified from 126.1, 54.7, 78.2, and 126.2 kW to 131.1, 49.7, 77.2, and 127.6 kW. Meanwhile, the corresponding temperatures are corrected in Figure 9. The final results of the modified HEN-ORC and the comparison with the literature are shown in Table 12, where W_{net} of the modified HEN-ORC is 46.3 kW in this work. It is 8%, 8%, and 5% higher than the ones in Desai and Bandyopadhyay,[44] Chen et al.,[45] and Watanapanich et al.,[18] where the conventional hydrocarbons (n-pentane, isohexane) are adopted as the working fluids to optimize the HEN-ORC system. The main difference between the proposed one and the above benchmark approaches is that no working fluid is specified in the proposed HEN-ORC model. Thus, the integrated design of the HEN-ORC system with new working fluids can be achieved. Besides, Desai and Bandyopadhyay[44] adopted a graphical approach. Chen et al.[45] and Watanapanich et al.[18] both adopted sequential approaches, where the utility consumption was first targeted and then the power output of ORC was optimized.

Two case studies show that the developed design strategy can achieve more economical and efficient results. This is because the working fluid candidates in this study contain 430,000 new HFOs, creating much more feasible solutions than the literature. Based on the quantitative heat-to-work model, the ideal HEN-ORC system can be optimized without a specific working fluid, and only new HFOs satisfying the optimized thermodynamic properties are further screened inversely. Compared with the literature, more power output is generated and lower TAC can be achieved for two case studies.

TABLE 10 Optimization results of Case 2.

	Results	Value
ORC	T^{ph} (K)	381.75
	T^{eva} (K)	449.95
	T^{sh} (K)	497.15
	T^{con} (K)	353.15
	T^{ds} (K)	395.75
	$F^{wf}Cp^{ph}$ (kW/K)	2.735
	$F^{wf}Cp^{sh}$ (kW/K)	2.671
	$F^{wf}Cp^{ds}$ (kW/K)	2.519
	$F^{wf}\Delta h^{eva}$ (kW)	54.7
	$F^{wf}\Delta h^{con}$ (kW)	213.4
	W_{net} (kW)	47.2
HEN	Q^{hu} (kW)	244.1
	Q^{cu} (kW)	126.2

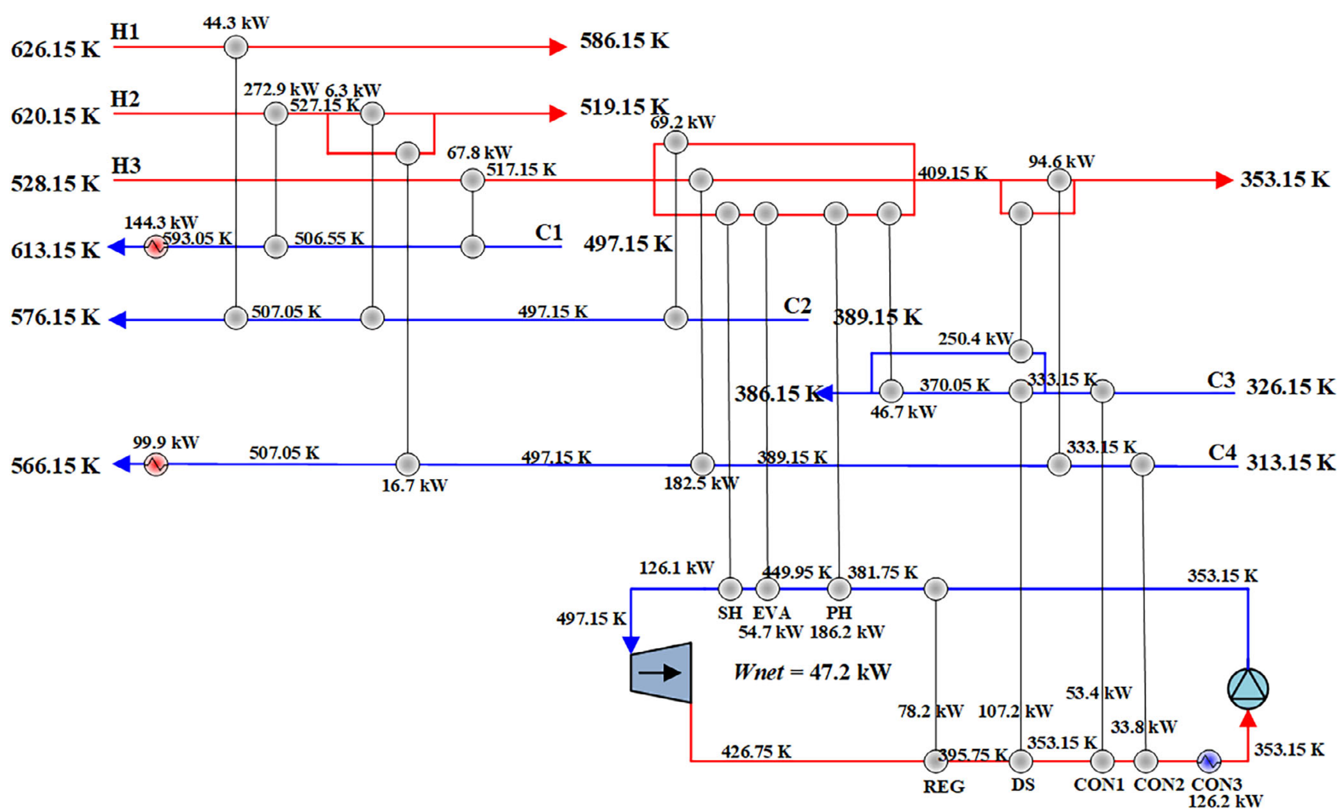
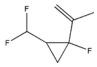
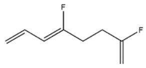
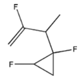
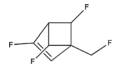
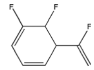


FIGURE 8 The ideal HEN-ORC results for Case 2.

TABLE 11 The 5 HFOs with the smallest error for Case 2.

Optimal HFOs	1	2	3	4	5
Molecule					
SMILES	CC(=C)C1(F)CC1C(F)F	FC(=C)CCC(F) = CC=C	CC(C(F) = C)C1(F)CC1F	FCC12C=C(F)C(C1F)C2F	FC1C(C=CC=C1F)C(F) = C
T^c (K)	455.95	490.25	476.85	450.35	459.15
$\tilde{F}^{wf}$ (kmol/s)	0.0089	0.0097	0.0104	0.0083	0.0084
$\tilde{Cp}^{ph}$ (kJ/kmol·K)	304.8	279.5	238.7	315.8	309.5
$\tilde{Cp}^{sh}$ (kJ/kmol·K)	311.1	281.4	296.5	355.0	336.5
$\tilde{Cp}^{ds}$ (kJ/kmol·K)	270.9	206.5	233.2	266.4	265.8
$\Delta\tilde{h}^{eva}$ (kJ/kmol)	5564.5	5365.6	5736.1	6188.8	7311.3
$\Delta\tilde{h}^{con}$ (kJ/kmol)	24055.9	21344.3	20525.4	25052.2	24724.1
$\tilde{F}^{wf}\tilde{Cp}^{ph}$ (kW/K)	2.721	2.723	2.481	2.627	2.592
$\tilde{F}^{wf}\tilde{Cp}^{sh}$ (kW/K)	2.778	2.741	3.081	2.953	2.819
$\tilde{F}^{wf}\tilde{Cp}^{ds}$ (kW/K)	2.419	2.012	2.423	2.216	2.227
$\tilde{F}^{wf}\Delta\tilde{h}^{eva}$ (kW)	49.7	52.3	59.6	51.5	61.2
$\tilde{F}^{wf}\Delta\tilde{h}^{con}$ (kW)	214.8	207.9	213.3	208.4	207.1
Φ	0.113	0.123	0.128	0.129	0.131

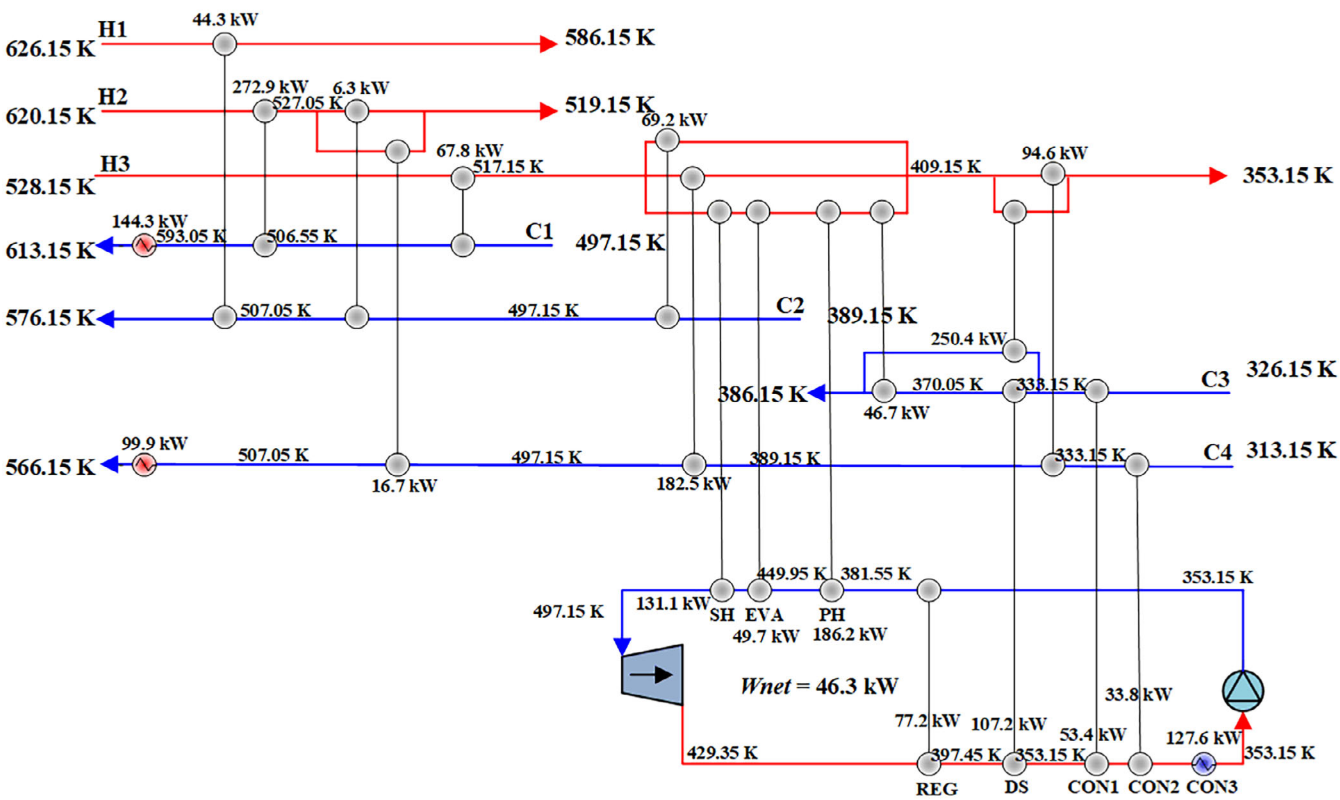
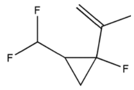


FIGURE 9 The modified HEN-ORC structure for Case 2.

TABLE 12 The result comparisons with the literature for Case 2.

Results	This work	Desai and Bandyopadhyay[44]	Chen et al.[45]	Watanapanich et al.[18]
Working fluids		n-Pentane	Isohexane	n-Pentane
T^{ph} (K)	381.55	354.15	354.45	376.55
T^{eva} (K)	449.95	459.65	466.15	454.45
T^{sh} (K)	497.15	/	/	/
T^{con} (K)	353.15	353.15	353.15	353.15
T^{ds} (K)	398.95	409.05	411.55	337.75
Q^{hu} (kW)	244.1	244.1	244.1	244.1
Q^{cu} (kW)	127.6	129.8	129.9	128.3
W_{net} (kW)	46.3	42.7	42.8	44.3

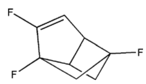
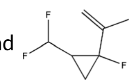
8 | OUTLOOK

The availability and advantages of the proposed two-stage method are demonstrated by the above Case 1 and Case 2. However, there are still areas for future research regarding the integrated design of HEN-ORC systems with working fluids.

1. Generative models, such as variational autoencoders[46] (VAEs) and quantum computing-assisted molecule generation framework,[47] can be applied for generating new HFOs. In the proposed method, working fluids are screened out from a database. Although the database contains 430,000 HFOs, there is still potential for further discovery.
2. Quantum computing algorithms,[48,49] particularly those based on quantum annealing, have shown promise in solving complex optimization problems more efficiently than classical methods. The proposed method could be applied in large-scale problems using quantum computing.
3. The accuracy and uncertainties of the thermodynamic property prediction model should be further considered. Potential working fluids may be overlooked due to inaccuracies in thermodynamic property predictions.
4. An integrated assessment framework that combines multiple sustainability metrics, such as GWP, ozone depletion potential (ODP), and ecological toxicity, should be developed to provide a holistic evaluation of the ORC-HEN system's environmental performance.
5. The chemical feasibility and stability of working fluids should be considered.

9 | CONCLUSION

Most of the general ORC working fluids will be phased out because of their higher GWP. It is urgent to develop new environmentally friendly and economical working fluids. HFOs with low GWP are expected to be widely used in ORC for waste heat recovery. Integrated design of HEN-ORC and working fluids is an effective way to improve the system performance, but it is challenging because of the complexity of the model. In this study, a new two-stage reverse design strategy from the HEN-ORC system to the HFO working fluid is proposed. A quantitative EOS-free heat-to-work model of ORC is developed, circumventing the nonlinearities associated with incorporating EOS or thermodynamic property prediction models. In the first stage, the optimal HEN-ORC structure and the thermodynamic properties of the hypothetical working fluid can be obtained by a MINLP problem. To find the best HFO candidate, whose properties match the hypothetical working fluid, two GC-ANN models are developed to predict the thermodynamic properties of the novel HFO candidates. One HFO working fluid database (Database #1) is established as the HFO candidate pool, and one hydrocarbon working fluid database (Database #2) is constructed to provide physical property data for training GC-ANN models. In the second stage, potential HFOs with properties closest to the optimized results are identified from Database #1.

The proposed design strategy is applied to two cases, where HEN-ORC systems and novel HFO working fluids are designed and compared with the literature results  and  are selected as novel working fluids for the two cases, respectively, which shows better economy and higher power output. The net power output of Case 1 and Case 2 is 5%–69% higher than the literature, and the total annual cost of Case 1 is 12%–22% higher than the literature.

NOMENCLATURE

Subscript

0	standard state
i	heat stream
j	cold stream
loss	loss
net	Net
r	relative
wf	working fluid

Superscript

abs	absorption process
c	critical state
com	compression process
con	condensation process
cu	cold utility
ds	cooling process
eva	evaporation process
ex	exchanger
exp	expansion process
ise	isentropic
iso	isothermal
liq	liquid
nis	non-isothermal
ph	preheat process
pump	Pump
reg	regeneration process
rej	rejection process
sat	saturated state
sh	superheating process
tur	turbine
vap	vapor
wh	waste heat

Parameters

C	unit cost
FCp	heat capacity flow rates
TIN	inlet temperatures
TOUT	outlet temperatures
α	cost index for turbines and pumps
β	area cost index
η	equipment efficiency
ω	acentric factor

Variables

Cp	heat capacity
F	flow rate
P	pressure
T	temperature
$\overline{T}$	mean temperature
W	power output
Δh	unit enthalpy change
Φ	error

AUTHOR CONTRIBUTIONS

Xiaodong Hong: Methodology; software; data curation; formal analysis; and writing – original draft. **Xuan Dong**: Methodology; software; data curation; formal analysis; and writing – original draft. **Zuwei Liao**: Conceptualization; funding acquisition; supervision; and writing – review and editing. **Jingyuan Sun**: project administration and writing – review and editing. **Jingdai Wang**: Supervision; funding acquisition; and resources. **Yongrong Yang**: Supervision; funding acquisition; and resources.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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